

INF10025 - TUTORIAL 03 COMPLETE ANSWERS

Detailed Solutions & Explanations | Week 3: Parameters, Action Queries, CrossTab, SQL & Import

Tài liệu Hướng dẫn Giải Chi tiết Bài tập Tutorial 3 (Week 3)

Bao gồm toàn bộ đáp án chính xác và lời giải thích chuyên sâu cho các chủ đề: Parameter Queries, Action Queries (Update, Delete, Make Table, Append), CrossTab Query, SQL, và Import dữ liệu từ tệp CSV/Excel.

1. Parameters (Truy vấn Tham số)

a) In the context of Access Queries, what is a Parameter?

Đáp án: A Parameter is a temporary named storage location in an Access query that prompts the user to enter specific criteria values at runtime when the query is executed.

(Giải thích: Tham số đóng vai trò là biến tạm thời, giúp tái sử dụng truy vấn động mà không cần tạo nhiều câu truy vấn tĩnh trùng lặp.)

b) Can a Parameter Name contain spaces?

Đáp án: YES. Parameter names in MS Access can contain spaces (e.g., [Type the beginning date:] or [Find Year]).

c) What is the relationship between the Parameter name and the prompt shown to end users when the query is executed?

Đáp án: The text enclosed inside the square brackets [Parameter Name] in the query design grid directly serves as the exact label/prompt text displayed inside the 'Enter Parameter Value' dialog box.

d) Assume that the text within square brackets below are query parameters:

Criteria: Between [Type the beginning date:] And [Type the ending date:]

What will happen when this query is executed?

Đáp án: When executed, Access will prompt the user twice sequentially: first asking 'Type the beginning date:' and then 'Type the ending date:'. It will then retrieve all records whose Invoice Date falls inclusively between those two entered dates.

2. Update Queries (Truy vấn Cập nhật)

a - d) Update Queries: True or False Questions

Câu hỏi (Question)	Định ước	Giải thích chi tiết (Explanation)
a) An update query may update multiple records within a single table in a single update query?	TRUE	Update Query cho phép chỉnh sửa hàng loạt bản ghi khớp với điều kiện cùng một lúc.
b) An update query may update multiple fields in a single update query?	TRUE	Có thể điền dòng 'Update To' cho nhiều cột khác nhau trong cùng 1 truy vấn.
c) An update query may use Parameters?	TRUE	Có thể kết hợp tham số dạng [Enter new value] ở dòng Update To hoặc Criteria.
d) An update query may also delete records?	FALSE	Update Query chỉ thay đổi giá trị trường dữ liệu, KHÔNG thể xóa bỏ bản ghi (đó là vai trò của Delete Query).

Update Query Example 1: Update To: 6.5 for TMDB_SCORE where RATINGCODE = 'PG' AND RELYEAR < 1980

- a) Output when executed:** Sets the TMDB_SCORE to exactly 6.5 for all movies rated 'PG' that were released before the year 1980.
- b) How to reverse if incorrect:** There is **NO Undo** for Action Queries! The easiest way to reverse it is to restore the database from a backup, or manually run another Update Query if previous score values were backed up.

Update Query Example 2: Update To: [tmdb_score]+1 for TMDB_SCORE where RATINGCODE = 'PG' AND RELYEAR < 1980

- c) Output when executed:** Increments (adds 1 point to) the existing TMDB_SCORE for all 'PG' movies released before 1980.
- d) Could such a query be reversed? YES.** You can easily reverse this by running another Update Query on the same records with Update To: [tmdb_score]-1.

3. Delete Queries (Truy vấn Xóa)

Query Design Grid Condition	Will records be deleted?	Detailed Explanation
a) Delete: Where Criteria: <i>(blank)</i>	NO	Không có điều kiện cụ thể được thiết lập nên không có bản ghi nào bị xóa.
b) Delete: Where Criteria: True	YES (ALL)	Điều kiện True luôn đúng với mọi hàng Xóa VĨNH VIỄN TẤT CẢ bản ghi trong bảng MOVIE!
c) Delete: Where Criteria: False	NO	Điều kiện False không bao giờ đúng Không có bản ghi nào bị xóa.
d) Delete: Where Criteria: <90 (RUNTIME) AND 'MA' (RATINGCODE)	YES (Filtered)	Chỉ xóa các phim có thời lượng dưới 90 phút ĐỒNG THỜI mang mã phân loại 'MA'.

4. Make Table / Append Queries & Data Integrity

a) Make Table Query Execution: Table Name: long_movies, Criteria: RUNTIME >= 150

Đáp án: Extracts MOVIEID, TITLE, and RUNTIME for all movies running 150 minutes or longer and creates a BRAND NEW table named long_movies to store them.

b) Copying 'Sales' to 'SalesCopy': How long until data is out of date?

Đáp án: IMMEDIATELY. As soon as any new transaction, update, or deletion occurs in the original Sales table, SalesCopy becomes outdated because it is a static, non-updated duplicate.

c) Data Integrity problem after correcting CustId 113 to 131 in Sales:

Đáp án: Loss of Data Integrity / Multiple Versions of the Truth. The corrected CustId 131 exists in Sales, but SalesCopy still holds the invalid CustId 113. This inconsistency causes conflicting reports depending on which table is queried.

d) How long until a saved Select Query 'SalesQueryFemale' becomes out of date?

Đáp án: NEVER. A query is a saved dynamic request (definition), not a static table copy. Every time SalesQueryFemale is executed, it fetches live, up-to-date data directly from the underlying Sales table.

5. CrossTab Query & SQL

CrossTab Query Purpose & Output: Row Heading: RELYEAR, Column Heading: COLOUR_CODE, Value: Count(MOVIEID)

Đáp án: Summarizes data in a two-dimensional grid format (like an Excel Pivot Table). It displays release years (RELYEAR) down the rows, color codes (COLOUR_CODE) across the columns, and calculates the total count of movies for each year-color combination.

SQL Questions: a) What is SQL? b) What has SQL got to do with Access Select Queries?

a) What is SQL: SQL (Structured Query Language) is the standardized ANSI/ISO database language used to query, define, and manipulate relational database management systems.

b) Relationship with Access: MS Access Query Design Grid (GUI) is simply a visual editor that automatically generates standard SQL code behind the scenes. Users can view or edit this code directly in SQL View.

6. Importing Data (CSV, Delimiters & Text Qualifiers)

a - e) CSV & File Import Basics

- **a) CSV File:** Comma-Separated Values file — plain text format storing tabular data separated by commas.
- **b) Delimiter:** A specific character (comma, tab, semicolon) separating individual field values.
- **c) Text Qualifier:** A character (usually double quotes ") wrapping text fields so embedded delimiters (like commas in addresses) aren't mistaken as column splits.
- **d) Renaming .CSV to .TXT:** File content remains unchanged; default OS application association changes, but Access can still import it.
- **e) Application opening .CSV:** Microsoft Excel opens by default due to Windows file association rules.

f) File Analysis: "Fred Bloggs"/"M"/230.00/"0493 900800"/"1 Red Rd, Hawthorn, 3122"

- **Delimiter character:** / (Forward Slash)
- **Text Qualifier character:** " (Double Quote)

g) Altered File without Text Qualifier & Comma Delimiter: Fred Bloggs,M,230.00,0493 900800,1 Red Rd, Hawthorn, 3122

CRITICAL IMPORT PROBLEM: The address string '1 Red Rd, Hawthorn, 3122' contains commas inside the text. Without text qualifiers, Access will misinterpret those internal commas as column delimiters! This splits a single address into 3 separate extra fields, causing severe column misalignment and schema errors during import.